Explainable AI for Early ICU Deterioration Prediction

Team members:

- Anuja Thuraiyur Jayakumar (25203657)
- Bageri Manjunath Ruthvik Gowda (25205410)

Introduction

Clinical Deterioration in ICU Patients:

- Intensive Care Units (ICUs) manage patients who may experience rapid clinical deterioration due to severe illness.
- Critical events include:
  - Respiratory failure
  - Septic shock
  - Cardiac arrest
  - Mortality
- Early prediction can support timely intervention and improve patient outcomes.
- Machine learning has shown promise in clinical deterioration prediction (Churpek et al., 2016).

Motivation

Why This Project?

- Traditional early warning systems may miss complex patterns.
- Electronic Health Records contain large volumes of patient data.
- Machine learning can identify high-risk patients earlier.
- Explainable AI improves transparency and trust.
- Supports AI-assisted clinical decision making (Tonekaboni et al., 2019).
- Increasing availability of Electronic Health Records (EHRs) enables predictive healthcare analytics.

Dataset

MIMIC-IV Critical Care Dataset

- Large publicly available database containing de-identified health data from intensive care patients.

Dataset Contents

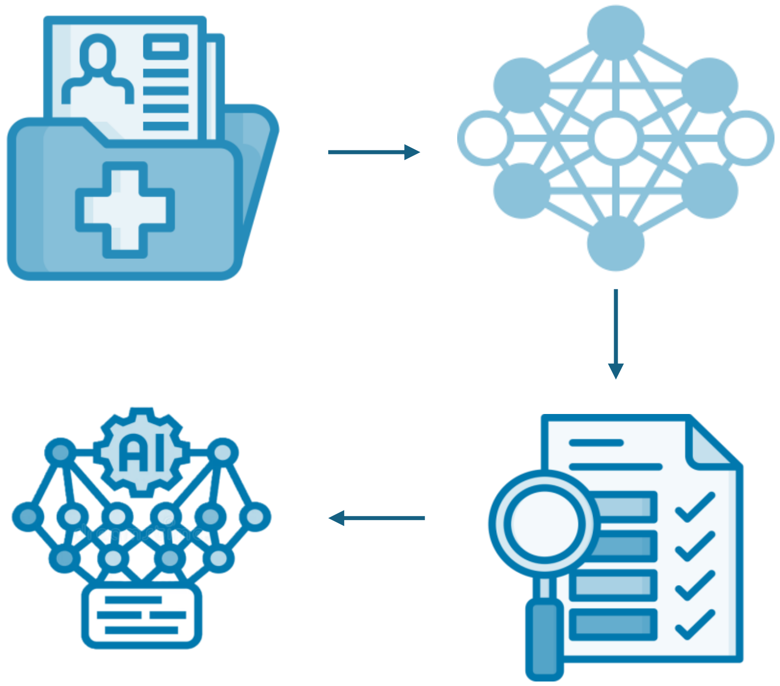
- Patient demographics
- Vital signs (heart rate, blood pressure, oxygen saturation, etc.)
- Laboratory measurements
- Diagnoses
- Procedures
- Medication records

Why MIMIC-IV?

- Widely used in healthcare AI research
- Large-scale real-world clinical dataset
- Suitable for predictive modelling

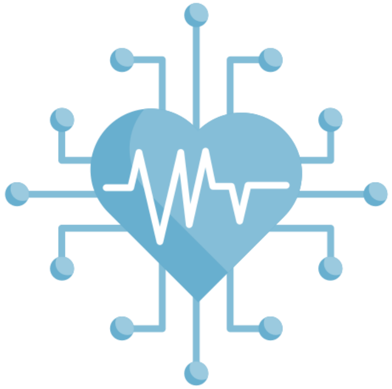
Proposed Methodology

- **Data Extraction from MIMIC-IV**
- **Data Preprocessing**
- **Feature Engineering**
- **Model Development**
  - Logistic Regression
  - Random Forest
  - XG Boost
- **Model Evaluation**
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - ROC-AUC
- **Explainability Analysis (SHAP)**



Expected Outcomes

- Predictive model for ICU deterioration.
- Comparison of machine learning approaches.
- Identification of key risk factors.
- Explainable AI analysis.
- Insights for clinical decision support systems.
- Identification of the most influential clinical features associated with patient deterioration.



Can machine learning models accurately predict clinical deterioration in ICU patients while providing interpretable explanations for their predictions?

References

- **Johnson, A. E. W., Bulgarelli, L., Shen, L., et al. (2023).**
MIMIC-IV, a freely accessible electronic health record dataset.
Scientific Data, 10(1), 1.
https://www.nature.com/articles/s41597-022-01899-x
- **Churpek, M. M., Yuen, T. C., Winslow, C., Meltzer, D. O., Kattan, M. W., & Edelson, D. P. (2016).**
Multicenter Comparison of Machine Learning Methods and Conventional Regression for Predicting Clinical Deterioration on the Wards.
Critical Care Medicine, 44(2), 368–374.
https://pubmed.ncbi.nlm.nih.gov/26771782/
- **Tonekaboni, S., Joshi, S., McCradden, M. D., & Goldenberg, A. (2019).**
What Clinicians Want: Contextualizing Explainable Machine Learning for Clinical End Use.
Proceedings of Machine Learning Research, 106, 359–380.
https://proceedings.mlr.press/v106/tonekaboni19a.html
- **Shickel, B., Tighe, P. J., Bihorac, A., & Rashidi, P. (2018).**
Deep EHR: A Survey of Recent Advances in Deep Learning Techniques for Electronic Health Record Analysis.
IEEE Journal of Biomedical and Health Informatics, 22(5), 1589–1604.
https://pubmed.ncbi.nlm.nih.gov/29989977/